Fishing for Privacy: Biological Confounds and the Limits of Membership Inference in Gene-Expression VAEs

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Abstract

Variational autoencoders trained on gene-expression data memorize rare biological subpopulations, and those are the samples where a membership inference attack (MIA) is most dangerous. A few patients with an unusual disease subtype, or a small cluster of rare cells, can be individually identifiable from model behavior. Can such memorization be selectively removed without retraining the model, and without destroying its utility? That requires fixing the measurement first. MIA protocols that draw matched negatives from different biological classes than the forget set can conflate subtype identity with training membership, and this confound affects general-purpose attacks (threshold, likelihood ratio, k-NN) along with the distance-based methods used in recent transcriptomics benchmarks. A trained MLP attacker on model-internal features, evaluated against a multi-seed retrain baseline under the identical matched-negative protocol, recovers a smaller memorization signal that sits almost entirely in biologically coherent subpopulations; within-subtype matching corroborates this where enough held-out cells of the forget class exist. The three cohorts play different roles under this evaluation. On Tabula Muris and TCGA-BRCA the retrain reference meets or exceeds the baseline, so no membership signal is present to unlearn, which is itself informative. On PBMC-33k, where a genuine signal survives, no post-hoc method reliably reaches retrain-level privacy across seeds and forget-set sizes. Methods that preserve utility produce no privacy improvement. The methods that do reduce the membership signal destroy the model through posterior collapse or Streisand effects (Hayes et al., 2024). The underlying cause is structural. A classifier forgets a class through its class-specific output weights, where forget and retain Fisher information barely overlap (cosine 0.018), but a VAE has no such separable weights. Every layer entangles forget and retain influence, from the encoder (0.345) to the shared decoder output (0.485), each 19 to 27× the classifier's, so a parameter-space edit disturbs the retain set wherever it acts. Proposition 1 formalizes the decoder case, and a cluster-conditional decoder that removes the output-layer overlap still fails to unlearn because the encoder overlap remains. If privacy actually matters, retrain.

1 Introduction

Deep learning models memorize training data, and for rare subpopulations this memorization may be necessary for generalization (Feldman, 2020). When those models are trained on sensitive biological information, an attacker can determine whether a specific sample was part of the training set through a membership inference attack (MIA). Gene expression profiles can reveal disease status, genetic predispositions, and other personal health information. Rare cell types and disease subtypes may be individually identifiable even within large, aggregated datasets.

Can a gene-expression VAE selectively forget a rare biological subpopulation without retraining and without destroying model utility? This question arises when all cells from a specific cluster or all patients of a molecular subtype must be removed from a trained model. Experiments here use cell-type clusters (PBMC-33k, Tabula Muris) and disease subtypes (TCGA-BRCA) as forget sets across two modalities (single-cell and bulk RNA-seq).

Before testing unlearning methods, the evaluation itself had to be fixed. MIA protocols that draw matched negatives from different biological classes than the forget set can conflate subtype identity with training membership. This affects general-purpose attacks (threshold (Yeom et al., 2018), likelihood ratio (Carlini et al., 2022), k-NN (Shokri et al., 2017)) as well as the distance-based methods used in recent transcriptomics benchmarks (Ozturk et al., 2026; Golob et al., 2026). Within-subtype matching, a trained MLP attacker on model-internal features, and a multi-seed retrain baseline together isolate the genuine memorization signal, which concentrates in biologically coherent subpopulations.

Under this corrected evaluation, PBMC-33k is the one cohort with a genuine membership signal, and there no post-hoc method reliably reaches retrain-level privacy; on Tabula Muris and TCGA-BRCA no signal is present to unlearn. Methods that preserve utility produce no privacy improvement. Methods that reduce the membership signal destroy the model through posterior collapse or Streisand effects, where the act of unlearning leaves a detectable trace that makes forget-set samples more distinguishable than before (Hayes et al., 2024). The cause is Fisher information overlap in the VAE's shared decoder, which prevents selective parameter perturbation.

Contributions.

1. A corrected evaluation protocol for MIA on biological generative models. Cross-class matched-negative protocols can conflate cell-type identity with training membership. The operative correction for the main results is a multi-seed retrain baseline evaluated under the identical matched-negative protocol together with a trained attacker on model-internal features; within-subtype matching corroborates it where enough held-out cells of the forget class exist. These isolate the genuine memorization signal, which concentrates in biologically coherent subpopulations. The confound magnitude varies by dataset and is documented per-dataset in Section 5.

2. A systematic evaluation of eleven unlearning approaches (nine approximate methods plus training-time synthetic augmentation and representation alignment) on PBMC-33k, with representative methods carried to Tabula Muris and TCGA-BRCA. The unlearning failure is established on PBMC-33k, the one cohort with a measurable membership signal, where methods either fail to reduce the signal or reduce it but destroy model utility. On scattered forget sets memorization is negligible and unlearning is unnecessary. On Tabula Muris and TCGA-BRCA the retrain reference meets or exceeds the baseline, so no membership signal is present to unlearn under the tested attack suite.

3. A structural explanation via Fisher information overlap. A classifier has class-specific output weights that can be targeted for forgetting (forget-retain cosine 0.018), but the VAE has no comparably separable weights at any layer, with overlaps of 0.345 to 0.493 across encoder, bottleneck, and decoder (19 to 27× the classifier; 0.75 on TCGA-BRCA). A cluster-conditional decoder that zeroes the output-layer overlap still fails, confirming the constraint is not decoder-specific. Proposition 1 formalizes the decoder case for linear decoders, with dimensional scaling as a heuristic ($1-O(M/D)$ for the generative residual-variance factor against $1/\sqrt{C}$ for a classifier).

2 Background and threat model

Unlearning and MIA. Machine unlearning produces parameters θ' from a trained model, forget set $\mathcal{F}$, and retain set $\mathcal{R} = \mathcal{D} \setminus \mathcal{F}$ such that a post-hoc MIA cannot distinguish forget samples from never-seen samples (Cao & Yang, 2015; Bourtoule et al., 2021). Theoretical guarantees exist for restricted model classes (Ginart et al., 2019; Guo et al., 2020; Sekhari et al., 2021; Neel et al., 2021; Izzo et al., 2021) but depend on convexity or smoothness that deep generative models violate. MIA determines training-set membership via shadow models (Shokri et al., 2017) or likelihood ratios (Carlini et al., 2022). Privacy leakage is reported as membership inference advantage (adapted from Yeom et al. (2018)): advantage $= 2|\text{AUC}-0.5|$, direction-agnostic because an adversary can always flip predictions. Hayes et al. (2024) showed that inexact unlearning can leave forgotten samples more detectable (Streisand effect); Thudi et al. (2022) argued that without auditable definitions, unlearning claims are unverifiable.

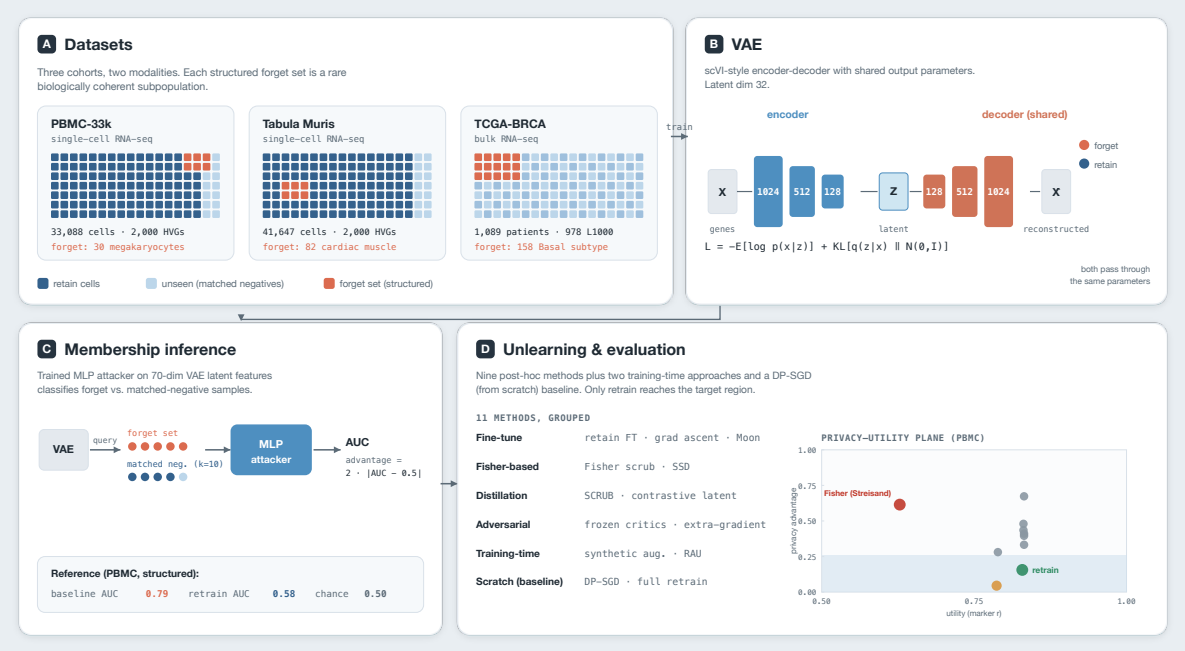


Figure 1: Evaluation pipeline. **(A)** Three cohorts across two modalities, each with a structured forget set that is a rare, biologically coherent subpopulation. **(B)** scVI-style VAE, latent dim 32. **(C)** A trained MLP attacker on 70-dim latent features classifies forget versus matched-negative samples; matched negatives are drawn within-subtype via k-nearest neighbors. **(D)** Eleven unlearning approaches (nine post-hoc, two training-time) plus DP-SGD on the privacy–utility plane; no post-hoc method reaches the target (advantage ≤ 0.258, the 95% CI upper bound of the retrain baseline), which only full retraining and from-scratch DP-SGD attain.

Threat model. The scope is subpopulation unlearning, the removal of entire biological groups (rare cell types, disease subtypes) rather than individual samples. Data poisoning, adversarial extraction, and reconstruction attacks are out of scope. The adversary has query access and trains a dedicated attacker on data they control, which is a stronger setting than off-the-shelf distance-based MIA such as GAN-leaks (Chen et al., 2020), Monte Carlo, or Mahalanobis distance. A model that faithfully captures subtype biology will by construction produce outputs close to real training members of the same subtype, yielding high MIA scores that reflect biological fidelity rather than memorization. The trained MLP attacker used here (70-dim features from the VAE latent space) learns which feature combinations are informative for membership specifically, as evidenced by its near-chance retrain advantage (0.16) compared to distance-based methods (advantage ≥ 0.43 on the retrain model; Table 3).

Related work. Fisher scrubbing (Golatkar et al., 2020), SSD (Foster et al., 2024), and SCRUB (Kurmanji et al., 2023) all assume forget-set influence concentrates in identifiable parameter subsets, an assumption that influence-based attributions are already known to strain in deep networks (Basu et al., 2021) and that Section 7 shows breaks down further in shared-decoder architectures. Nasr et al. (2018) and Chavdarova et al. (2019) provide the adversarial min-max and extragradient machinery used in extra-gradient co-training; Abadi et al. (2016) introduced DP-SGD, used here as a formal privacy baseline. Pawelczyk et al. (2025) found that approximate unlearning fails to remove data-poisoning attacks, a parallel negative result in the classifier setting. Moon et al. (2024) proposes feature unlearning for pre-trained generators by fine-tuning the decoder along a target-feature latent direction, evaluated here as a post-hoc method (Table 2). Cheng et al. (2026) studies retain-forget entanglement in classifiers via Wasserstein regularization, without a Fisher-overlap account; the present work locates the entanglement structurally in the shared decoder output of a generative model and formalizes it (Proposition 1), showing it defeats unlearning precisely where a classifier's class-specific outputs avoid it. On biological data, Walker et al. (2024) showed single-cell count matrices leak

Table 1: Dataset and split summary. Matched negatives are the k-nearest unseen samples to the structured forget set in baseline latent space ($k = 10$). TCGA-BRCA data from Ozturk et al. (2026), VST-normalized, 978 LINCS L1000 landmark genes (Subramanian et al., 2017).

	PBMC-33k	Tabula Muris	TCGA-BRCA
Type	scRNA-seq	scRNA-seq	Bulk RNA-seq
Samples	33,088 cells	41,647 cells	1,089 patients
Genes	2,000 HVGs	2,000 HVGs	978 L1000
Train / Unseen	28,124 / 4,964	35,399 / 6,248	925 / 164
Structured forget	Cluster 13 (30 megakaryocytes)	Cluster 33 (82 cardiac muscle)	Basal subtype (158 patients)
Forget fraction	0.1%	0.2%	17%
Patient-level forget	N/A	N/A	5, 10, 20 patients
Matched negatives	194	137	121

private information via eQTL linkage. Golob et al. (2026) introduced scMAMA-MIA for scRNA synthetic data and observed per-cell-type vulnerability but set it aside as out of scope; Ozturk et al. (2026) reported $\rho = 0.92$ between differential expression recovery and MIA vulnerability across 11 generative models on TCGA. Both use distance-based MIA on synthetic outputs and do not stratify by subtype. Lopez et al. (2018) developed scVI, the architecture tested here.

3 Datasets and setup

Figure 1 summarizes the full evaluation pipeline. This section describes the datasets and the forget-set construction, and Section 5 develops the within-subtype evaluation protocol.

The PBMC-33k dataset consists of 33,088 peripheral blood mononuclear cells from 10x Genomics, preprocessed with Scanpy (Wolf et al., 2018), each represented by 2,000 highly variable gene counts. The structured forget set is cluster 13, which contains 30 rare megakaryocytes. The Tabula Muris dataset (The Tabula Muris Consortium, 2018) has 41,647 cells from 12 mouse tissues with 35 Leiden clusters; the structured forget set is cluster 33 (82 cardiac muscle cells). The TCGA-BRCA dataset (Cancer Genome Atlas Network, 2012) consists of 1,089 breast cancer patients with PAM50 molecular subtypes (LumA 52%, LumB 20%, Basal 17%, Her2 8%, Normal 4%), preprocessed by Ozturk et al. (2026) for the CAMDA 2025 Health Privacy Challenge. Each sample corresponds to one patient, enabling patient-level forget sets. Table 1 gives a full comparison.

To control for biological confounds in MIA evaluation, matched negatives were selected as the unseen cells closest to the forget set in the baseline model's latent space (k-NN with $k = 10$). Without this control, an attacker could distinguish forget from unseen cells by cell type alone rather than by memorization. A within-subtype holdout variant restricts matched negatives to the same biological group as the forget set. On PBMC, this drops baseline AUC from 0.769 toward chance (5 unseen megakaryocytes, an underpowered control detailed in Section 5); on TCGA-BRCA, baseline and retrain AUCs converge to 0.576 (30 unseen Basal patients). Section 5 quantifies this confound across datasets.

4 Methods

4.1 VAE architecture

The VAE follows scVI design principles (Lopez et al., 2018). The encoder maps input genes (2,000 for PBMC/Tabula Muris; 978 for TCGA-BRCA) through layers $[1024, 512, 128]$ to latent mean μ and log-variance $\log \sigma^2$ with latent dimension 32. The decoder reverses this through $[128, 512, 1024]$ to a softmax output across genes. Layer normalization and dropout (0.1) follow each hidden layer. Because the input data are log-normalized (Section 3), the reconstruction term is mean squared error (Gaussian observation model

on the transformed counts); the training objective is the negative ELBO, $\mathcal{L}(\theta, \phi; x) = -\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] + \mathrm{KL}(q_\phi(z|x) \| p(z))$.

4.2 Unlearning methods

Nine post-hoc approximate unlearning methods were tested, alongside DP-SGD (Abadi et al., 2016) as a formal privacy baseline that trains from scratch on the retain set. Full hyperparameters and pseudocode are in the released code; the summary below names each approach and cites its origin. **Retain-only fine-tuning** continues VAE training on $\mathcal{R} = \mathcal{D} \setminus \mathcal{F}$. **Gradient ascent** maximizes forget-set loss then fine-tunes on the retain set. **Frozen critics** freeze pre-trained MIA attackers and update the VAE to minimize critic success. **Fisher information scrubbing** (Golatkar et al., 2020) updates $\theta \leftarrow \theta + \alpha(F + \lambda I)^{-1} \nabla_\theta \mathcal{L}_\mathcal{F}$ with the diagonal Fisher information F on the forget set. **SSD** (Foster et al., 2024) dampens parameters multiplicatively in proportion to forget-set Fisher importance. **SCRUB** (Kurmanji et al., 2023) uses teacher-student distillation that matches on retain data and diverges on forget data. **Moon feature unlearning** (Moon et al., 2024) builds a forget-versus-retain latent direction and fine-tunes the decoder so forget-set latents reconstruct with that direction removed while retain reconstructions are preserved, leaving the encoder fixed. **Contrastive latent unlearning** pushes forget-set posteriors toward the prior $\mathcal{N}(0, I)$ while preserving retain-set representations. **Extra-gradient co-training** frames unlearning as a min-max game,

$$\min_\theta \max_\psi \mathcal{L}_{\mathrm{VAE}}(\theta; \mathcal{R}) - \lambda \cdot \mathcal{L}_{\mathrm{att}}(\theta, \psi; \mathcal{F}), \tag{1}$$

and uses the two-step extragradient update of Chavdarova et al. (2019) with two-time-scale updates (TTUR, attacker LR 10× lower than VAE LR), three co-trained critics, and 50 epochs (stopping before equilibrium). Hyperparameters for each method were chosen by sweep on PBMC-33k; results below report multi-seed evaluation at the chosen settings.

4.3 Attack suite

The canonical attacker is a trained MLP on 70-dimensional features (69 VAE latent statistics plus a k-NN distance to the retain set, $k = 5$; spectral normalization, two hidden layers of 256, dropout 0.3). A single attacker is trained on the baseline model's forget-versus-matched-negative features, where the membership signal is strongest, and then applied unchanged to every target model, including each unlearned checkpoint and the retrain reference. This fixed decision rule is if anything more lenient than an attacker refit to each model, so the finding that every post-hoc method still fails against it is conservative; the same attacker scores the retrain reference near chance, which calibrates the measurement and shows the rule is tuned to membership rather than to a per-model boundary refit to noise. A refit control confirms this directly. An attacker trained from scratch on each model's own features, scored out-of-fold across the five retrain seeds and averaged over 20 random fold splits, reaches advantage 0.13 (AUC 0.567, nested 95% CI $[0.05, 0.22]$ on advantage), consistent with the transferred attacker's 0.16 whose CI it overlaps, while the same refit protocol recovers advantage 0.60 (AUC 0.80) on the baseline model. The dedicated adversary extracts no more from the retrain model than the fixed rule does, and its power on the baseline rules out saturation at $n = 30$, so the near-chance retrain score reflects the model rather than a weak attacker (Appendix D). Three additional attack families provide robustness checks. **Threshold attacks** apply a threshold to reconstruction loss, KL divergence, or ELBO individually (Yeom et al., 2018) and require only black-box access. **Likelihood-ratio** compares per-sample loss under the target model against the retrain reference (Carlini et al., 2022). **k-NN latent** classifies membership by distance to known training samples in latent space ($k = 10$) (Shokri et al., 2017).

The evaluation target is a 5-seed retrain baseline (seeds 42–46), yielding fresh-attacker advantage 0.16 with an exact nested 95% CI $[0.08, 0.26]$ (AUC = 0.578) that combines cross-seed variance with within-eval bootstrap uncertainty on the ≈ 224-sample matched-negative pool. Every method carries the same nested CI, and each is compared to retrain by a nested bootstrap of the advantage difference rather than a test on seed means alone. A permutation null with no membership signal present sets the advantage estimator's floor at 0.091 (95% $[0.003, 0.253]$), so the retrain reference itself sits only marginally above chance, consistent with the biology confound. The per-method advantage is the mean across seeds of the folded per-seed value

$2|\mathrm{AUC}_s - 0.5|$; for high-variance methods this exceeds $2|\overline{\mathrm{AUC}} - 0.5|$ (extra-gradient reports 0.281 against 0.134 from the mean AUC of 0.433), which is why the permutation floor calibrates the estimator and why extra-gradient's advantage CI $[0.20, 0.37]$ overlapping that floor is the honest reading of its result.

5 The biology confound

On TCGA-BRCA (1,089 breast cancer patients, bulk RNA-seq), the structured forget set is 158 Basal-subtype patients (17% of training). With standard cross-subtype matching, baseline AUC = 0.803 and retrain AUC = 0.830. The retrain model, which never saw the Basal patients, is better at detecting them than the baseline. All six tested unlearning methods converge to the retrain floor (AUC $\approx$ 0.830).

Within-subtype matching (restricting matched negatives to 30 unseen Basal patients) resolves the cross-subtype confound. Baseline and retrain advantages both reach ≈ 0.15 and become indistinguishable from each other. The entire cross-subtype gap was driven by Basal identity. The residual within-subtype advantage of 0.15 is not zero, so the attacker still detects something, but whatever it detects is equally present in the retrain model that never saw the Basal patients, and is therefore not a membership signal. Patient-level experiments at $n_f \in \{5, 10, 20\}$ show the same pattern, with baseline and retrain advantages overlapping within their CIs at every size and no consistent gap.

The confound replicates on PBMC. Restricting matched negatives to the 5 unseen megakaryocytes drops the baseline AUC toward chance, but with only 5 negatives the estimate is unstable (separate runs give 0.53 to 0.60) and the confound fraction it implies carries a bootstrap CI of $[-0.37, 0.98]$, so this control is not statistically resolved. The load-bearing replication is PBMC cluster 7 (CD14+ monocytes, 80 unseen cells), where cross-cluster AUC = 0.665 falls to within-cluster AUC = 0.527 on a larger holdout, so most of the cross-cluster above-chance signal is subtype identity rather than memorization. Tabula Muris cluster 28 (67 unseen) shows no detectable per-sample memorization under either matching strategy. Table 3 shows the confound at the attack level, where high-assumption attacks (k-NN, likelihood ratio) achieve advantage ≥ 0.73 on the retrain model because they detect biological structure regardless of training history.

A second rare cluster shows the same confound. PBMC cluster 12 (49 cells) has baseline advantage 0.75 (AUC 0.88), yet a model retrained without those cells still reaches advantage 0.64 (AUC 0.82, three seeds, standard deviation 0.01). The genuine membership gap is only ≈ 0.11 in advantage against 0.43 for cluster 13, so a biology confound carries most of the apparent cluster-12 signal. Because the retrain reference is itself confound-inflated, matching it is a weak target on cluster 12, and the informative result is method brittleness. Extra-gradient at $\lambda = 10$, marginal on cluster 13, drives the attacker to chance on every cluster-12 seed (AUC 0.50, advantage 0.03) at a utility cost (marker correlation falls from 0.83 to 0.76); Fisher scrubbing collapses the posterior (marker correlation 0.21); retain-only fine-tuning leaves advantage at baseline (0.73). The extra-gradient configuration tuned on cluster 13 does not transfer, so the one method competitive there is not robust to the choice of forget set.

Figure 2 in Appendix E visualizes the cross-subtype vs. within-subtype comparison and decomposes the cross-subtype AUC into biology and memorization components.

The confound is consistent with two recent benchmarks that report high MIA scores on scRNA and bulk RNA using distance-based MIA on synthetic outputs. Ozturk et al. (2026) report $\rho = 0.92$ between differential expression recovery and MIA vulnerability across 11 generative models (TCGA-COMBINED, $n = 11$ so the correlation is suggestive), and Golob et al. (2026) observe cell-type-dependent vulnerability in scMAMA-MIA but set the dependence aside as out of scope. Distance-based methods on synthetic outputs cannot distinguish biological signal preservation from individual-level memorization; a trained MLP attacker on model-internal features can, as its lower retrain advantage (0.16) compared to baseline (0.58) on PBMC shows.

Table 2: All methods on PBMC-33k structured forget set (cluster 13, $n = 30$), from the per-seed evaluation with a fresh canonical attacker. Advantage carries an exact nested 95% CI combining cross-seed and within-eval (∼224-sample) bootstrap uncertainty. Status is set by comparing the per-seed advantage to the retrain 95% CI upper bound (0.258) and by a nested bootstrap of the method-minus-retrain advantage difference. Every post-hoc method exceeds the bound, and every post-hoc difference CI excludes zero except extra-gradient, whose difference from retrain is not resolved while its point advantage still exceeds the bound. Under a permutation null with no membership signal present, the advantage estimator itself reports 0.091 (95% $[0.003, 0.253]$), so values near this floor are not distinguishable from chance. †Single-checkpoint entries carry within-eval CIs only and are excluded from the difference test. AUCs here are multi-seed means; the single-run canonical attacker used in Sections 5 and 6.2 reports baseline 0.769 and retrain 0.481 on one fixed checkpoint pair.

Method	Seeds	AUC	Advantage [95% CI]	Marker r	Status
Baseline (no unlearning)	1	0.791	0.582	0.831	anchor
Retain-only fine-tune	5	0.666	0.333 [0.23, 0.43]	0.832	FAIL
Gradient ascent	5	0.698	0.396 [0.30, 0.49]	0.832	FAIL
SSD ($\alpha = 1.0$)	3	0.718	0.435 [0.31, 0.56]	0.831	FAIL
SCRUB ($\alpha_f = 1.0$)	3	0.706	0.411 [0.28, 0.54]	0.832	FAIL
Moon feature-unlearn	3	0.740	0.480 [0.36, 0.60]	0.831	FAIL
Contrastive latent ($\gamma = 1.0$)	3	0.164	0.673 [0.57, 0.75]	0.832	FAIL (Streisand)
Fisher scrubbing†	1	0.808	0.615 [0.45, 0.77]	–	FAIL (worse)
Frozen critics† ($\lambda = 10$)	1	0.992	0.984	–	FAIL
Extra-grad ($\lambda = 10$)	10	0.433	0.281 [0.20, 0.37]	0.789	FAIL (marginal)
DP-SGD ($\varepsilon = 10$)	3	0.478	0.045 [0.03, 0.18]	0.787	≈ retrain*
Full retrain	5	0.578	0.156 [0.08, 0.26]	0.829	TARGET

6 Unlearning results

6.1 Unlearning methods on structured forget set

Table 2 groups the nine post-hoc methods by failure mode alongside the DP-SGD baseline.

Utility-preserving, privacy-failing. Retain-only fine-tuning, gradient ascent, SSD, and SCRUB preserve utility (marker $r \geq 0.831$) but produce no useful privacy reduction, with advantages of 0.33 to 0.44. Thirty forget-set cells leave too small a gradient signal against the 28,094 retain cells. Each has a method-minus-retrain advantage difference whose nested 95% CI excludes zero (probability above retrain ≥ 0.99). SSD at $\alpha = 5.0$ (single seed) lowers advantage to 0.27, still above the retrain CI upper bound (0.26).

Detectable artifacts. Three methods make the model worse rather than better. Contrastive latent unlearning drops AUC to 0.164 (advantage = 0.673), a Streisand effect (Hayes et al., 2024) where the forced move to the prior is itself a detectable signature. Fisher scrubbing collapses the posterior (KL falls from 10.55 to 0.007) and advantage rises to 0.615. Frozen critics (AUC = 0.992†) exploit fixed decision boundaries without removing the underlying signal.

Adversarial and formal. Extra-gradient with $\lambda = 10$ (10 seeds) reduces mean advantage to 0.281 but with $\sigma_{\text{AUC}} = 0.142$; per-seed behavior spans over- and under-unlearning, and its advantage difference from retrain is not statistically resolved (nested 95% CI $[-0.01, 0.24]$ includes zero, probability above retrain 0.97), so it is the single method whose separation from retrain is not established, though its point advantage still exceeds the retrain CI upper bound (0.258). The method fails outright on Tabula Muris (Section 6.2) and, on a second PBMC forget set, the same $\lambda = 10$ drives the attacker to chance at a utility cost (advantage 0.03, Section 5) rather than reaching a stable target, so the configuration does not transfer across forget sets. DP-SGD at $\varepsilon = 10$ reaches advantage = 0.045, and its difference from retrain is not distinguishable from

Table 3: Attack suite results on PBMC-33k structured forget set. Advantage = $2|\text{AUC} - 0.5|$. Low-assumption attacks use only loss components; high-assumption attacks require a reference model or training set embeddings. The trained MLP (canonical attacker) is shown for comparison.

	Attack	Baseline	Retrain	EG λ=10	Fisher
Low	Recon. threshold	0.438	0.438	0.438	0.438
	KL threshold	0.841	0.733	0.192	0.272
	ELBO threshold	0.838	0.725	0.023	0.454
High	Likelihood ratio	0.839	0.839	0.734	0.744
	k-NN latent	0.986	0.971	0.897	0.927
	Trained MLP	0.538†	0.038†	0.036*†	0.628†

*Single seed; multi-seed mean advantage = 0.281. †Separate evaluation attacker.

zero (nested 95% CI $[-0.19, 0.04]$), but it is not unlearning; the forget set was never in training, and ELBO rises from 364 to 403 while marker r drops from 0.831 to 0.787. The * in Table 2 marks this distinction.

Fisher behaves differently across forget-set types. On scattered forget sets it achieves AUC = 0.499 (near chance) but from a baseline of 0.525, so the actual privacy gain is $\Delta = 0.026$; on structured sets it collapses the posterior and makes privacy worse. The one method that works on scattered sets fails where memorization is strongest (full ablation in Appendix L).

6.2 Attack diversity

Reconstruction-loss thresholding gives identical advantage (0.438) across all models because reconstruction quality is preserved, so the membership signal lives in the KL and latent geometry. ELBO and KL thresholds drop under extra-gradient (from 0.838 to 0.023 for ELBO), but Fisher's apparent KL drop ($0.841 \to 0.272$) is posterior collapse, not unlearning. High-assumption attacks (k-NN latent, likelihood ratio) keep advantage ≥ 0.73 even on retrain, so they detect cell-type identity, not membership. Only the trained MLP on internal features is near-chance on retrain.

Cross-dataset validation. On Tabula Muris, the retrain model has AUC = 0.944, exceeding the baseline (0.891). Since retrain never saw cluster 33, the attacker is detecting cardiac muscle biology rather than membership; TM does not measure membership leakage at the per-cell level (full table in Appendix J). Extra-gradient (0.874) and Fisher (0.946) therefore confirm rather than refute the biology-confound thesis (Section 5); no method moves the AUC below the retrain ceiling.

6.3 Constructive approaches

Two constructive approaches were tested to see whether the structural limit can be bypassed. **Synthetic augmentation at training time** trained VAEs on retain + bootstrap-resampled synthetic megakaryocytes (from 5 unseen seed cells with Gaussian noise). On PBMC with within-cluster matching, the augmented model achieves AUC = 0.82–1.00, worse than baseline (0.63); the model's representation of the megakaryocyte region is now shaped by the 5 seed cells, and the MIA detects the shift. Augmentation replaces one memorization pattern with another. On TCGA-BRCA the augmented model (0.581) matches baseline (0.578) and retrain (0.573), consistent with no patient-level memorization. **Representation alignment unlearning (RAU)** fine-tunes the baseline so forget-sample posteriors match the retrain model's with loss $\mathcal{L} = \text{ELBO}(\mathcal{R}) + \lambda \cdot \text{KL}(q_{\text{student}}(z|x_f) \| q_{\text{retrain}}(z|x_f))$. A sweep over $\lambda \in \{0.1, 1, 10, 100\}$ aligns posteriors (KL to retrain drops from 139 to 0.02) but pushes cross-cluster AUC to 0.10–0.13, a Streisand effect. The structural limit extends to representation space; any modification to forget-sample treatment, in parameter space (Fisher, SSD) or representation space (RAU, contrastive), creates a detectable trace.

Table 4: Utility metrics on PBMC-33k held-out set (4,964 cells). Marker r is the average Pearson correlation across 8 cell-type marker genes. Full per-gene results in Appendix A.

	ELBO ↓	KL	Marker r ↑	ARI ↑
Baseline	364.2	10.55	0.831	0.452
Retrain	365.6	9.69	0.829	0.488
Retain-FT	363.2	10.59	0.832	0.461
Grad-ascent	363.3	10.53	0.832	0.441
SSD ($\alpha = 1.0$)	363.9	10.40	0.831	0.449
SCRUB ($\alpha_f = 1.0$)	363.8	10.41	0.832	0.451
Contrastive ($\gamma = 1.0$)	364.0	10.38	0.832	0.458
Extra-grad ($\lambda = 10$, 10-seed)	403.7	4.70	0.789	0.508
Fisher	489.8	0.007	0.628	0.481
DP-SGD ($\varepsilon = 10$, 3-seed)	403.3	8.08	0.787	0.482

6.4 Utility evaluation

Utility on the held-out set (4,964 cells) is shown in Table 4. Five methods (retain-FT, gradient ascent, SSD, SCRUB, contrastive) preserve utility (ELBO $\approx$ 364, marker $r \geq 0.831$) but barely change the model, which is why they fail on privacy. Extra-gradient and DP-SGD degrade ELBO by $\sim$40 and marker r by $\sim$5%, the cost of any actual signal reduction. Fisher collapses the posterior entirely (ELBO = 490, marker $r = 0.628$, KL = 0.007). ARI is preserved throughout, so global cluster structure survives even when gene-level reconstruction degrades.

The privacy-utility plane (Figure 4 in Appendix K) shows the resulting pattern. The five utility-preserving methods cluster at excellent ELBO but high advantage; extra-gradient and DP-SGD trade utility for privacy but do not reach the retrain target; only retrain occupies the low-advantage, low-ELBO region.

7 Why parameter-space methods fail

Fisher information overlap. Parameter-space unlearning (Fisher scrubbing, SSD, SCRUB) modifies parameters identified as important for the forget set via the diagonal Fisher $F_i = \mathbb{E}[(\partial\mathcal{L}/\partial\theta_i)^2]$. For this to work, forget-set Fisher $F^{\mathcal{F}}$ and retain-set Fisher $F^{\mathcal{R}}$ must be separable; cosine similarity measures that separability, with cosine near 1 meaning no parameter-space perturbation can reduce forget-set loss without proportionally affecting the retain set. Table 5 reports the diagonal Fisher computed on the forget set (30 megakaryocytes) and retain set (28,094 cells) from the baseline PBMC model, alongside a linear classifier (logistic regression on frozen VAE latents, 14 Leiden clusters) trained on the same data.

A classifier can forget a class by editing the weights that produce that class's output score. The forget set concentrates its Fisher information on the 32 weights that produce the cluster-13 score, while the retain set spreads across the weights for the other 13 class scores, giving a near-orthogonal forget-retain cosine of 0.018 at the output layer. The VAE has no comparable set of weights to target. Forget and retain influence are entangled at every layer, from the encoder (0.345) through the bottleneck (0.428) and decoder hidden layer (0.351) to the decoder mean output (0.493), each 19 to 27$\times$ the classifier's 0.018 (Table 5). Whichever parameters a method perturbs, it perturbs weights where forget and retain influence are aligned. The decoder mean output is the layer the Fisher methods actually edit, whose 1,024-dimensional hidden representation is shared across all 2,000 output genes, so perturbing any column affects reconstruction of every cell type. Its overlap gives the sharpest single contrast, 0.485 for the mean weight alone (Table 5 lists 0.493 for the pooled decoder-output category, which includes the vestigial dispersion head, whose cos = 0.996 is a damping artifact that receives zero gradients under MSE) against the classifier's 0.018, a 27$\times$ gap in the layer where unlearning perturbs. The global cosine (0.209) understates the constraint because forget and retain influence concentrate in different high-magnitude layers, so pooling all parameters dilutes the per-layer alignment. At the gene level, 73% of genes (1,453/2,000) have Fisher cosine > 0.5 with median 0.857.

Table 5: Fisher information overlap between forget and retain sets. Cosine similarity measures how entangled forget-set influence is with retain-set influence in parameter space. Higher cosine means less room for selective parameter perturbation. The classifier's low overlap shows that class-specific output weights allow targeted forgetting; the VAE offers no comparably separable weights at any layer. The effective rank ratio (F) reports the forget-set Fisher's effective rank (the exponential of its normalized-diagonal entropy) as a fraction of the layer's parameter count, so small values indicate influence concentrated on few coordinates.

Model	Layer category	Parameters	Cosine sim.	Eff. rank ratio (F)
VAE	Encoder	2,642,816	0.345	0.030
VAE	Bottleneck	8,256	0.428	0.033
VAE	Decoder hidden	598,912	0.351	0.041
VAE	Decoder output	4,100,000	0.493	0.006
VAE	**Global**	**7,349,984**	**0.209**	0.010
Classifier	**Output layer**	**462**	**0.018**	0.037

A structural bound on Fisher overlap. The pattern above admits a closed-form explanation for linear decoders.

Proposition 1 *Let $f(z) = Wz + b$ with $W \in \mathbb{R}^{D\times H}$ and squared-error loss. If residuals $e_d = x_d - f(z)_d$ are independent of latent activations z_h, the diagonal Fisher factorizes as $F_{dh} = 4\,\mathbb{E}[e_d^2]\,\mathbb{E}[z_h^2]$, and*

$$\cos(F^{\mathcal{F}}, F^{\mathcal{R}}) = \cos(\sigma^{\mathcal{F}}, \sigma^{\mathcal{R}}) \cdot \cos(\nu^{\mathcal{F}}, \nu^{\mathcal{R}}) \tag{2}$$

where $\sigma^S \in \mathbb{R}^D$ has entries $\sigma_d^S = \mathbb{E}_{x\sim S}[e_d^2]$ (residual variance per output dimension) and $\nu^S \in \mathbb{R}^H$ has entries $\nu_h^S = \mathbb{E}_{x\sim S}[z_h^2]$ (latent second moment).

Proof sketch. $\partial\mathcal{L}/\partial W_{dh} = -2e_d z_h$, so $F_{dh} = 4\,\mathbb{E}[e_d^2 z_h^2] = 4\,\sigma_d\nu_h$ under independence. Then $\langle F^{\mathcal{F}}, F^{\mathcal{R}}\rangle = 16\,\langle\sigma^{\mathcal{F}}, \sigma^{\mathcal{R}}\rangle\,\langle\nu^{\mathcal{F}}, \nu^{\mathcal{R}}\rangle$ and $\|F^S\| = 4\,\|\sigma^S\|\,\|\nu^S\|$. Dividing gives (2). □

Remark. A perturbation weighted by forget-set Fisher importance ($\delta\theta_i^2 \propto F_i^{\mathcal{F}}$) disturbs retain-set curvature by $\cos(F^{\mathcal{F}}, F^{\mathcal{R}}) \cdot \|F^{\mathcal{R}}\|/\|F^{\mathcal{F}}\|$. Near-zero cosine (classifier) admits perturbation directions that primarily affect the forget set; cosine = 0.485 in the shared decoder output (VAE) does not.

Corollary 2 (Dimensional scaling) *Under the conditions of Proposition 1:*

1. *If $D - M$ of the D output dimensions satisfy $\sigma_d^{\mathcal{F}} = \sigma_d^{\mathcal{R}}$ and the remaining M dimensions have $\sigma_d^S \le V\bar{\sigma}$ for all S, then*

$$\cos(\sigma^{\mathcal{F}}, \sigma^{\mathcal{R}}) \;\ge\; \frac{D - M}{D - M + MV^2}.$$

2. *For a single-class forget set where $\sigma^{\mathcal{F}}$ is supported on a single coordinate k, $\cos(\sigma^{\mathcal{F}}, \sigma^{\mathcal{R}}) = \sigma_k^{\mathcal{R}}/\|\sigma^{\mathcal{R}}\|$, which equals $1/\sqrt{C}$ for balanced C-class residuals.*

Proof in Appendix F. On PBMC, $\cos(\nu^{\mathcal{F}}, \nu^{\mathcal{R}}) = 0.60$ because forget and retain activations share the decoder's 1,024-dim hidden output (KL regularization constrains z but not h). Roughly 50–100 marker genes of $D = 2{,}000$ differ in forget-set residual variance; Corollary 2(i) with $M = 100$, $V = 3$ gives a lower bound of 0.68, which the data-direct $\cos(\sigma^{\mathcal{F}}, \sigma^{\mathcal{R}}) = 0.85$ satisfies. For the classifier, part (ii) predicts $1/\sqrt{14} \approx 0.27$ as an upper bound; the measured 0.018 is lower because the forget-class gradient is more concentrated than a point mass at 95% accuracy. The Fisher matrix is approximately rank-1 (leading singular value explains 94–98% of Frobenius norm), and the Fisher-marginal factorization $\cos(\sigma) \cdot \cos(\nu) = 0.50$ overshoots the measured 0.485 by 3% (the same product from data-direct factors is 0.53); Appendix G reports the full breakdown.

Controls. A deep MLP classifier ($2{,}000 \to [512, 128] \to 14$, 1.09M parameters) has shared-hidden cosine 0.53 (matching VAE encoder 0.35, decoder hidden 0.35) and class-specific output 0.006 (matching linear probe 0.018), so overlap depends on parameter sharing rather than capacity. Reducing VAE latent dimension to $z = 8$ raises global cosine to 0.35 from 0.21; on TCGA-BRCA (978 genes, 17% forget fraction) it reaches 0.75, as Corollary 2 predicts (Appendix H).

Implication. Because no layer offers a set of weights that affects the forget set alone, a parameter-space edit cannot reduce forget-set loss without proportionally disturbing the retain set, wherever in the network it acts. This is the structural reason the parameter-space methods fail rather than a proof that no method can. A cluster-conditional decoder tests the claim directly. Cluster-specific output columns drive the decoder's forget-retain overlap to near zero ($\cos \approx 10^{-9}$), the separable weights a classifier enjoys, yet the model still does not unlearn (advantage 0.76 to 0.71), because the membership signal is read from the encoder, where 64 of the 70 attacker features originate, and the encoder overlap (0.62 in this model) is untouched (Appendix I). Separating one set of weights while the rest stay entangled does not help, which is what the whole-network reading predicts and the decoder-only story does not. Classifier unlearning works because the class-specific output weights are the separable target the VAE lacks at every depth. The overlap argument bounds the achievable tradeoff, the absence of a clean selective direction, not the specific failure mode observed; the most aggressive methods overshoot into posterior collapse, the same constraint expressed through over-perturbation. The mechanism plausibly extends to other generative models with shared output parameters, though only scVI-style VAEs are tested here, and it does not by itself account for adversarial methods such as extra-gradient, whose failure is one of optimization and transfer rather than Fisher locality. Proposition 1 is a linearized heuristic for the decoder case of the nonlinear softmax model, and the encoder overlap is analogous but not formalized here. The analysis uses the diagonal Fisher, which Kunstner et al. (2019) showed can misrepresent the true Fisher; the concern is limited because these methods use the diagonal internally (Golatkar et al., 2020; Foster et al., 2024; Kurmanji et al., 2023), and the log-Fisher correlation ($r = 0.71$) and per-gene pattern (73% > 0.5, Appendix M) corroborate at different granularities.

Gradient projection and feature fine-tuning. Two recent strategies target this entanglement from opposite ends. Moon feature unlearning edits the decoder along a forget-versus-retain latent direction while a retain reconstruction term holds utility fixed; run here, it keeps utility at baseline level (marker $r = 0.831$, ELBO 364) yet leaves advantage at 0.480, because the bulk of the attacker's signal comes from the 64 latent-moment features read off the untouched encoder. Its parameter-space counterpart, Cheng et al. (2026), pairs a two-phase augmented-Lagrangian increase of the forget loss with a Wasserstein-2-regularized projection that steers the update off directions degrading the retain distribution. Such a projection needs a direction that raises forget loss while sparing retain, which is available when the forget and retain gradients are close to orthogonal, as a classifier's class-specific outputs are. In the shared decoder the two are far more aligned than in a classifier, so a separating direction is correspondingly scarcer, and the method is conjectured to struggle in this regime. This is a prediction rather than a test. Cheng's projection acts on mean-gradient directions, whereas the 0.485-versus-0.018 contrast is a diagonal-Fisher second-moment quantity used here as a proxy, and instantiating the method on the VAE would settle it directly.

8 Discussion

Unlearning as a detectable operation. Contrastive, Fisher scrubbing, frozen critics, and RAU all produce behavioral changes the attacker picks up. Contrastive creates an unnatural latent location, Fisher collapses the posterior, frozen critics drive the VAE toward critic-specific blind spots, and RAU leaves a representational gap between student and retrain model. Synthetic augmentation substitutes one detectable signature for another. This is the Streisand effect (Golatkar et al., 2020; Hayes et al., 2024). Verification is a related problem. The only ground truth is the retrain model, which the data controller is trying to avoid running; DP-SGD offers a formal guarantee but requires training from scratch.

Limitations. Privacy guarantees are empirical (except DP-SGD). Tabula Muris is confounded by tissue-of-origin signals; within-subtype PBMC uses only 5 unseen megakaryocytes; TCGA-BRCA shows no within-

subtype gap, so it cannot test methods that need one. Only VAE architectures were tested. The size ablation (Appendix B) conflates size with heterogeneity.

9 Conclusion

Structured subpopulations are memorized. Scattered samples are not. On PBMC-33k, the one cohort with a measurable signal, no post-hoc method reliably reaches retrain-level privacy across seeds and sizes; on Tabula Muris and TCGA-BRCA no signal is present to unlearn. The VAE entangles forget and retain influence at every layer (19 to 27× a classifier's targetable output weights), so no parameter-space edit has a clean target, a structural limit of the architecture that a cluster-conditional decoder confirms; the corrected evaluation protocol is what makes the limit visible. Among the methods tested, the dependable options avoid post-hoc unlearning altogether. Full retraining reaches the target by construction, and DP training from scratch matches it with a formal guarantee at a utility cost, while no post-hoc method reliably does.

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A Full utility metrics

Table 6: Per-gene marker correlation (r) on PBMC-33k held-out set for 8 cell-type marker genes. Fisher shows severe degradation on NK markers (GNLY, NKG7), consistent with posterior collapse disrupting specific decoder pathways.

Gene	Baseline	Retrain	Retain-FT	Grad-asc.	Extra-grad	Fisher
CD3D	0.767	0.761	0.770	0.769	0.602	0.601
CD3E	0.645	0.642	0.644	0.648	0.532	0.552
MS4A1	0.809	0.812	0.810	0.808	0.693	0.761
CD79A	0.900	0.900	0.901	0.901	0.727	0.815
CD14	0.798	0.797	0.800	0.797	0.767	0.644
LYZ	0.901	0.901	0.900	0.899	0.875	0.835
NKG7	0.930	0.928	0.928	0.929	0.698	0.515
GNLY	0.900	0.894	0.903	0.903	0.636	0.304
Mean	0.831	0.829	0.832	0.832	0.691	0.628

Extra-gradient preserves reconstruction quality for most genes but degrades T-cell (CD3D/E) and NK-cell (NKG7/GNLY) markers; the per-gene values shown are from a single seed and the 10-seed mean marker r (0.789) is higher than this seed. Fisher is worst on GNLY ($r = 0.304$ vs. baseline 0.900) and NKG7 ($r = 0.515$), consistent with posterior collapse disrupting cell-type-specific decoder mappings. Retain-only fine-tuning, gradient ascent, SSD, SCRUB, and contrastive latent unlearning are all indistinguishable from baseline on all genes (marker $r \geq 0.831$), consistent with these methods making only minimal parameter changes. DP-SGD at $\varepsilon = 10$ shows a similar per-gene pattern to extra-gradient, with moderate degradation on T-cell and NK markers.

B Forget set size ablation

Table 7: PBMC-33k structured size ablation for extra-gradient $\lambda = 10$ (3 seeds except $n = 30$ which uses the 10-seed evaluation from Table 2). Larger forget sets span multiple clusters and the method fails to unlearn them.

Size	Clusters	Mean AUC	95% CI	Advantage	Status
10	13 (subset)	0.263 ± 0.055	[0.200, 0.326]	0.474	Over-unlearn
30	13 (all)	0.429 ± 0.142	[0.327, 0.530]	0.282	FAIL
50	13 + 12	0.503 ± 0.109	[0.379, 0.626]	0.203	Borderline
100	13 + 12 + 11	0.611 ± 0.028	[0.580, 0.642]	0.222	Insufficient

At $n = 10$, extra-gradient over-unlearns (advantage = 0.474). At $n = 30$ (10 seeds), mean AUC is 0.429 with high variance. At $n = 50$, the mean AUC of 0.503 is near the retrain floor, and mean advantage = 0.203 falls within the nested retrain CI ($[0.082, 0.258]$) from the multi-seed reference. However, per-seed AUCs span 0.41 to 0.66, so individual seeds range from over-unlearning to under-unlearning; this is unreliable across seeds rather than reliably at target. At $n = 100$, the method under-unlearns. A separate scattered-set ablation ($n = 10$ to 500, 10 seeds) confirmed that scattered cells have low baseline memorization (AUC 0.52–0.63) and Fisher provides at most marginal improvement (Δ of -0.01 to 0.04).

C Matched negative validation

The default matched negatives (194 cells, k-NN from unseen set with $k = 10$) include cells from multiple clusters (98.5% non-megakaryocytes), which means the attacker could partially rely on cell-type differences rather than pure membership. A within-cluster control using only the 5 unseen cluster-13 cells drops the

baseline AUC toward chance (separate runs give point estimates of 0.53 and 0.60), with retrain and extragradient likewise near chance. With only 5 negatives the estimate is not statistically resolved, its implied confound fraction spanning a bootstrap CI of $[-0.37, 0.98]$, and retrain exceeds 0.5 here because both groups (30 forget, 5 unseen) were excluded from retraining and the attacker overfits to within-cluster heterogeneity. The load-bearing confound evidence is therefore the cluster-7 holdout (80 negatives) in Section 5, not this control, which is limited by the small number of held-out cells from rare clusters.

D Refit attacker control

The canonical attacker is trained once on the baseline model's features and applied unchanged to every target model. This transfer could in principle understate leakage if the fixed decision rule is too weak to find signal that a model-specific adversary would surface, which matters most for the retrain reference where the near-chance score underpins the no-membership reading. To rule this out, a fresh attacker (same architecture, 70-dim features, spectral normalization, two hidden layers of 256, dropout 0.3, 50 epochs) was trained directly on each model's own forget-versus-matched-negative features. Because training and scoring the same points in-sample would inflate the estimate through memorization of the 224 feature vectors, the reported AUC is out-of-fold from stratified 5-fold cross-validation, so no point is scored by an attacker that trained on it, and the whole cross-validation is repeated over 20 random fold splits with the attacker re-seeded per split and fold, so the estimate is not tied to one split. The baseline model serves as a positive control.

The refit protocol has ample power at $n = 30$. On the baseline model it recovers advantage 0.60 (AUC 0.80), close to the canonical baseline signal, so it is not saturating. Across the five retrain seeds it reaches mean advantage 0.13 (AUC 0.567 ± 0.015), with a nested 95% CI of $[0.51, 0.62]$ on AUC and $[0.05, 0.22]$ on advantage that combines cross-seed and out-of-fold bootstrap uncertainty (Table 8). The estimate is stable across fold splits, with a per-seed split-induced standard deviation of at most 0.011 in AUC. The nested advantage CI upper bound of 0.22 stays below the 0.258 success threshold, and the value is consistent with the transferred attacker's retrain advantage of 0.16 whose CI it overlaps. Granting the adversary a model-specific decision boundary on the retrain model extracts no more signal than the fixed rule, so the near-chance retrain score is a property of the model, not an artifact of a weak attacker. The in-sample analogue of the canonical estimator (80/20 train, scored on the full set, an upper bound that grants the adversary the in-sample advantage) gives the same picture, with mean retrain AUC 0.573.

Table 8: Refit attacker (trained per model, out-of-fold AUC averaged over 20 random stratified 5-fold splits) versus the transferred canonical attacker on the retrain reference. The refit protocol recovers strong signal on the baseline positive control and lands near chance on retrain, matching the transferred attacker. Per-seed split-induced SD is at most 0.011 in AUC. Advantage $= 2|\text{AUC} - 0.5|$. Canonical numbers 'outputs/p4/multiseed/retrain_refit.json'.

Model	AUC	Advantage
Baseline (positive control)	0.798	0.597
Retrain seed 42	0.561	0.121
Retrain seed 43	0.593	0.186
Retrain seed 44	0.561	0.122
Retrain seed 45	0.562	0.124
Retrain seed 46	0.558	0.116
Retrain (5-seed mean)	0.567 ± 0.015	0.134 ± 0.029
Retrain nested 95% CI	$[0.51, 0.62]$	$[0.05, 0.22]$
Transferred attacker (canonical)	0.578	0.156

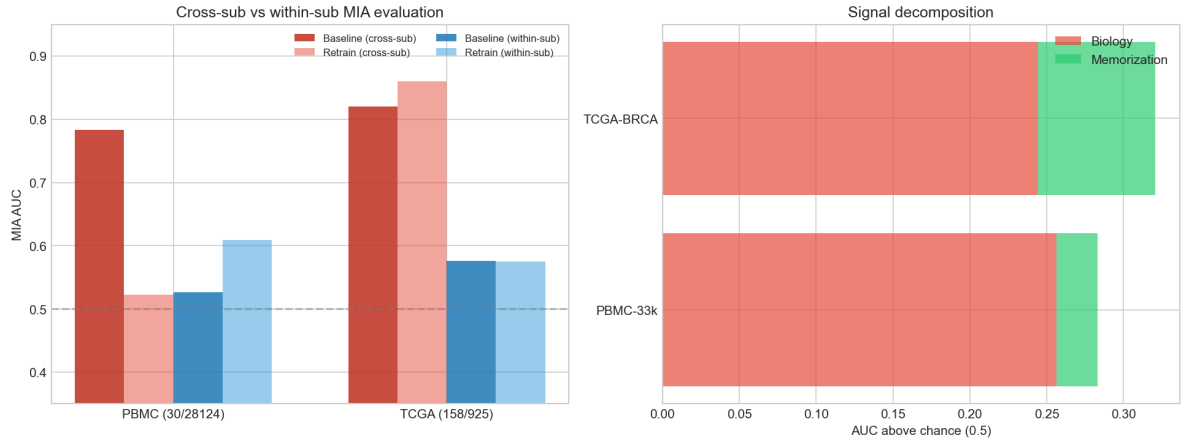


Figure 2: Biology confound across datasets. Left: cross-subtype vs. within-subtype MIA AUC for PBMC and TCGA-BRCA. With within-subtype matching, baseline and retrain AUCs converge. Right: decomposition of the cross-subtype AUC signal into biology (red) and memorization (green) components.

E Biology confound figure

F Corollary 2 proof

(i) $\langle \sigma^{\mathcal{F}}, \sigma^{\mathcal{R}} \rangle \geq (D-M)\bar{\sigma}^2$ from the shared dimensions, while $\|\sigma^S\|^2 \leq (D-M)\bar{\sigma}^2 + MV^2\bar{\sigma}^2$ for each set, giving the stated bound, which tends to 1 as $D/M \to \infty$. (ii) For $\sigma^{\mathcal{F}} = \sigma_k^{\mathcal{F}} \mathbf{e}_k$ the cosine reduces to $\sigma_k^{\mathcal{R}}/\|\sigma^{\mathcal{R}}\|$, which equals $1/\sqrt{C}$ for balanced C-class residuals. □

G Empirical verification of the factorization

The fc_mean Fisher matrix is approximately rank-1, with the leading singular value explaining 98% of the Frobenius norm for the forget set and 94% for the retain set, consistent with the outer-product form $F_{dh} \approx 4\sigma_d \nu_h$ from Proposition 1. The Fisher-marginal factorization $\cos(\sigma^{\mathcal{F}}, \sigma^{\mathcal{R}}) \cdot \cos(\nu^{\mathcal{F}}, \nu^{\mathcal{R}}) = 0.50$ overshoots the measured cosine of 0.485 by 3%. The residual error comes from the softmax output Jacobian, which couples output dimensions and departs from the element-wise independence assumed in Proposition 1. Under the per-sample estimator both factors track their data-direct counterparts closely, with Fisher-marginal ν (0.60) against the data-direct value (0.63) and Fisher-marginal σ (0.83) against the data-direct residual variance (0.85), and the product computed directly from data statistics (0.53) overshoots the measured cosine by 10%. The 27× gap in the shared output layer over the classifier (0.485 vs. 0.018) is consistent with the structural prediction.

H Fisher overlap across datasets

I Cluster-conditional decoder

A conditional VAE was trained in which the mean output layer receives the final hidden representation concatenated with a 14-dimensional cluster one-hot vector (adding 28K parameters). The fc_mean weight matrix then decomposes into shared columns (corresponding to the 1,024 hidden features) and cluster-specific columns (corresponding to the 14 one-hot inputs). The shared columns have cosine = 0.379, reflecting irreducible overlap from the shared hidden representation. The cluster-specific columns have cosine ≈ 0 (5×10^{-9}), achieving the same near-orthogonality as the classifier output weights. Fisher scrubbing on this conditional VAE was applied with the same hyperparameters as the standard experiments. The conditional VAE baseline shows higher memorization (advantage = 0.76) than the standard VAE (0.54), likely because

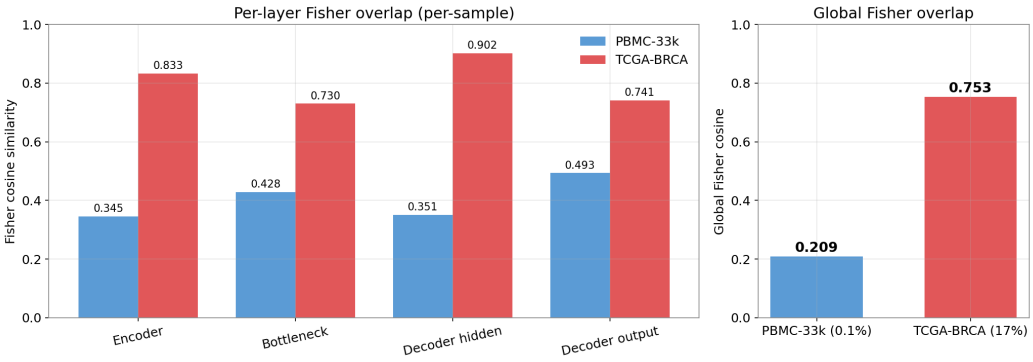


Figure 3: Fisher overlap on PBMC-33k (0.1% forget fraction) versus TCGA-BRCA (17%). Left: per-layer cosine. Right: global cosine. TCGA-BRCA overlap is uniformly higher, driven by the larger forget fraction and fewer output dimensions.

the cluster-specific columns give the model additional capacity to distinguish cluster 13. After scrubbing, advantage drops to 0.71, a negligible reduction. Fisher overlap persists throughout the shared network with encoder cosine = 0.62, bottleneck = 0.65, and decoder hidden = 0.51. Since the MIA attacker draws 64 of its 70 feature dimensions from encoder outputs, the high encoder overlap is the dominant factor. Routing cluster-specific information through dedicated output parameters does not reduce memorization already encoded in the shared encoder and hidden layers.

J Cross-dataset validation

Table 9: Cross-dataset comparison (PBMC-33k vs. Tabula Muris). PBMC values use the single-seed attacker (baseline = 0.769, retrain = 0.481; cf. Table 2 multi-seed where baseline = 0.791, retrain = 0.578). The Tabula Muris retrain AUC (0.944) exceeds the baseline (0.891), so the TM attacker is measuring cell-type identifiability, not membership. †Fisher fails on PBMC (gap 0.241 above multi-seed retrain) but is uninterpretable on TM where there is no membership signal to remove.

Method	PBMC		Tabula Muris		Gen.?
	AUC	Baseline	AUC	Baseline	
Retrain (structured)	0.481	0.769	0.944	0.891	–
EG $\lambda = 10$ (structured)	0.482	0.769	0.874	0.891	No
Fisher (structured)	0.814	0.769	0.946	0.891	PBMC fails†
Fisher (scattered)	0.499	0.525	0.568	0.411	–

K Privacy-utility plane

L Fisher unlearning on scattered forget sets

Fisher scrubbing behaves differently by forget-set structure (Table 10). Scattered forget sets are near-chance at baseline (AUC = 0.525), and Fisher brings advantage to $\Delta = 0.026$ above baseline, a marginal improvement of cells that were barely memorized. On structured sets, the posterior collapses and advantage rises from 0.538 to 0.628. Fisher cannot reduce the signal where the signal exists.

M Per-parameter Fisher scatter

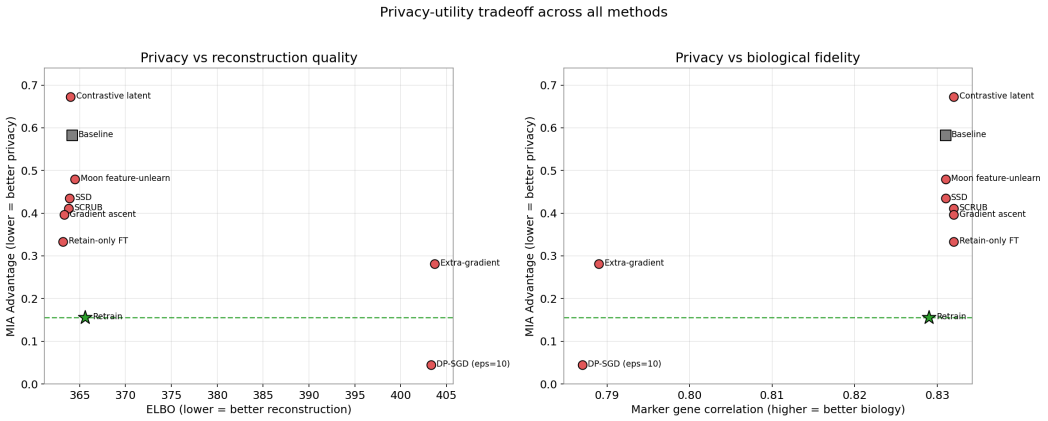


Figure 4: Privacy-utility tradeoff across all methods. Left: advantage vs. ELBO. Right: advantage vs. marker gene correlation. Methods that preserve utility (upper cluster) fail on privacy; methods that reduce advantage (DP-SGD, extra-gradient) pay a utility cost. Only retrain achieves both.

Table 10: Fisher unlearning on PBMC-33k by forget set type (3 seeds). Fisher reduces AUC on scattered sets, where baseline memorization is already weak. On structured sets it increases AUC through posterior collapse.

Forget type	Mean AUC	Std	Baseline AUC	Status
Structured	0.814	0.003	0.769	FAIL (worse)
Scattered	0.499	0.004	0.525	At chance

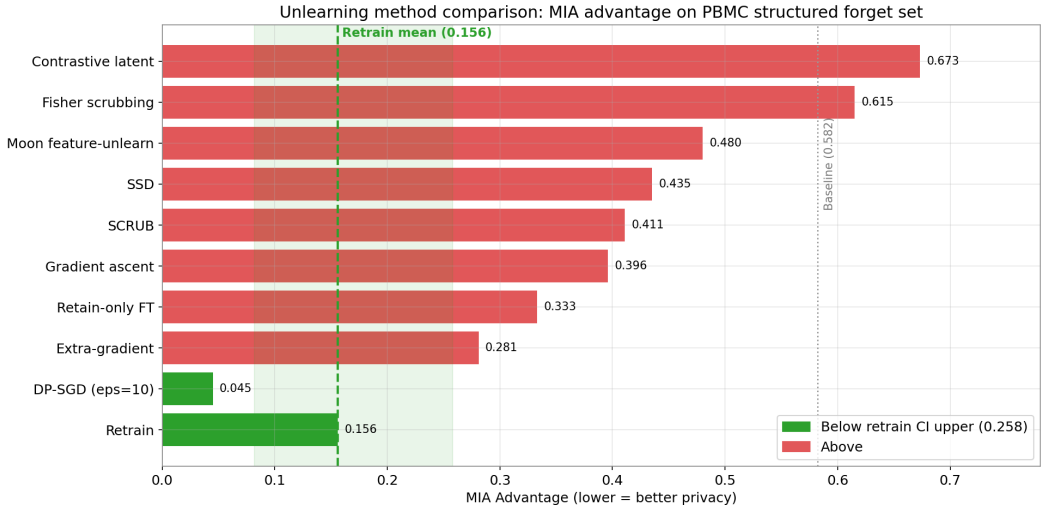


Figure 5: MIA advantage by method on PBMC-33k structured forget set ($n = 30$). The dashed line marks the 5-seed retrain advantage mean (0.156) with shaded nested 95% CI $[0.082, 0.258]$. DP-SGD's advantage (0.045) is not distinguishable from retrain but trains from scratch. No post-hoc unlearning method's advantage-difference CI reaches zero.

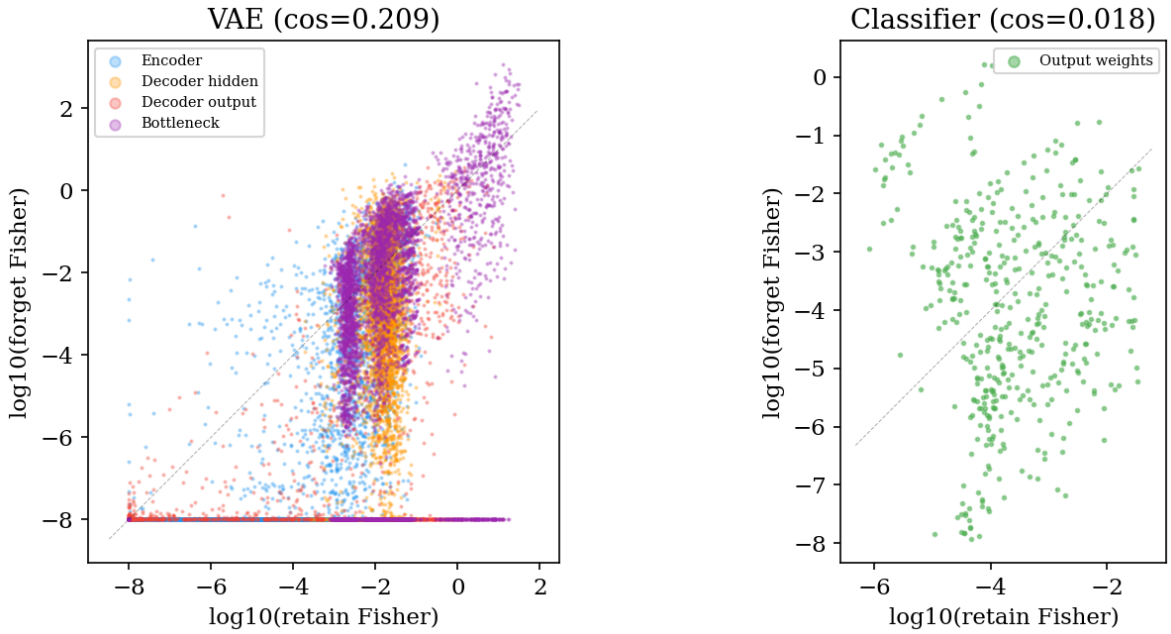


Figure 6: Per-parameter Fisher magnitude (log scale) for forget vs. retain sets. Left: VAE parameters are correlated (log-Fisher $r = 0.71$), confirming that the same parameters matter for both sets. Right: classifier parameters show no correlation (cos = 0.018), with forget-set Fisher concentrated in a few class-specific weights.